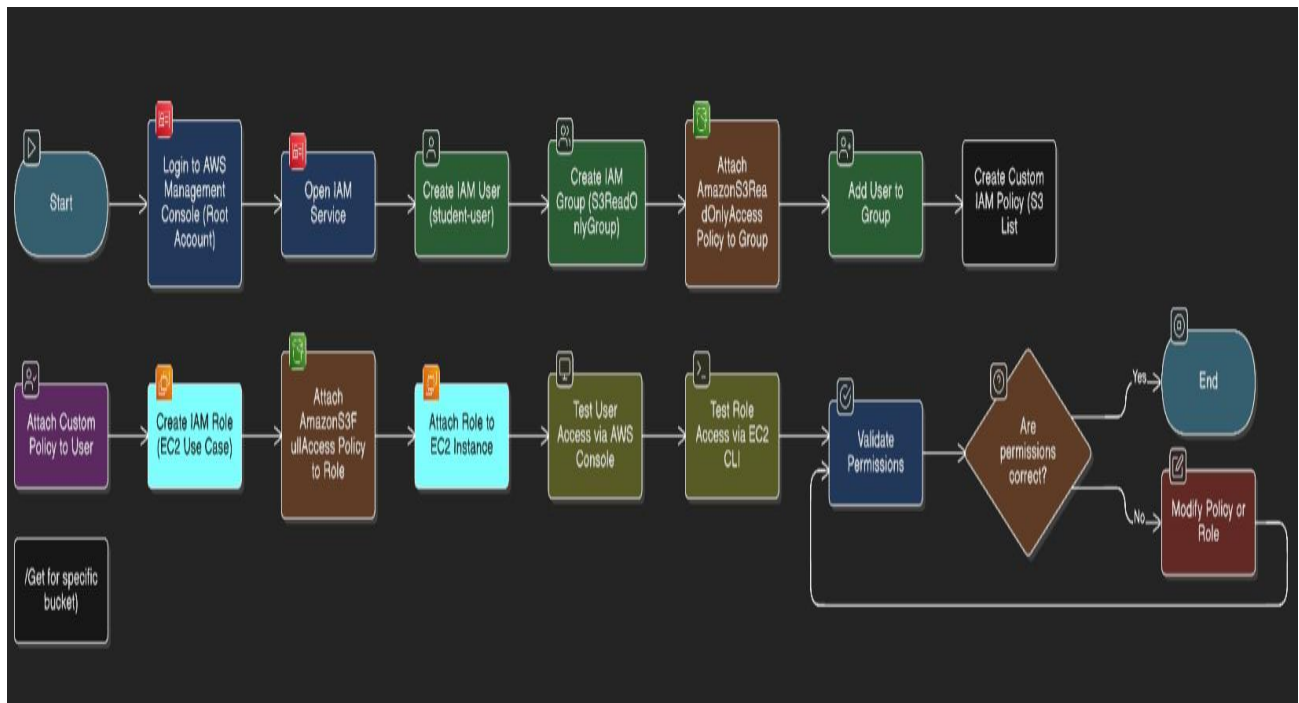


EXPERIMENT 5 : Identity and Access Management

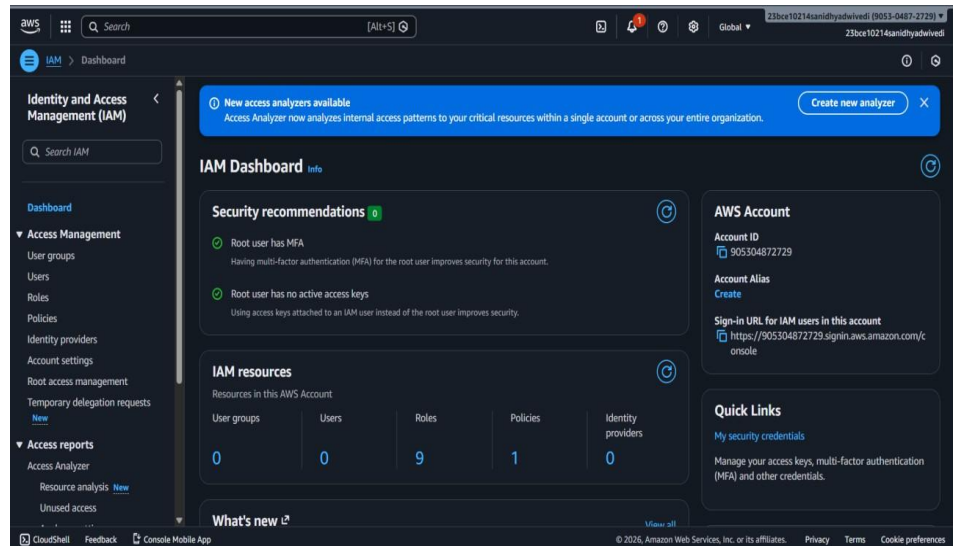


Step 1: Login to AWS Management Console

1. Go to: <https://console.aws.amazon.com>
2. Log in using your **root or IAM admin account**.

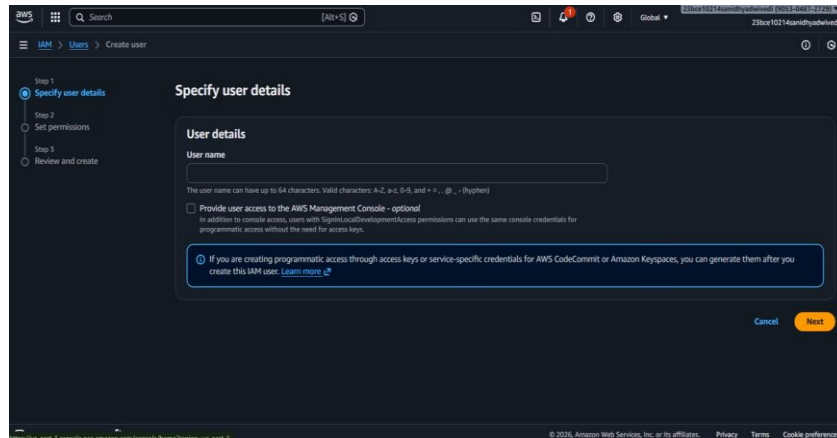
Step 2: Open IAM Service

1. In the AWS Console search bar → type **IAM**
2. Click **Identity and Access Management (IAM)**



Step 3: Create an IAM User

1. IAM Dashboard → Click **Users** → **Create user**
2. Enter:
 - **User name:** student-user
3. Select access type:
 - AWS Management Console access
 - Programmatic access (if CLI is needed)
4. Set Console password:
 - Choose **Autogenerated password** OR **Custom password**
 - Check *User must create a new password at next sign-in*
5. Click **Next**



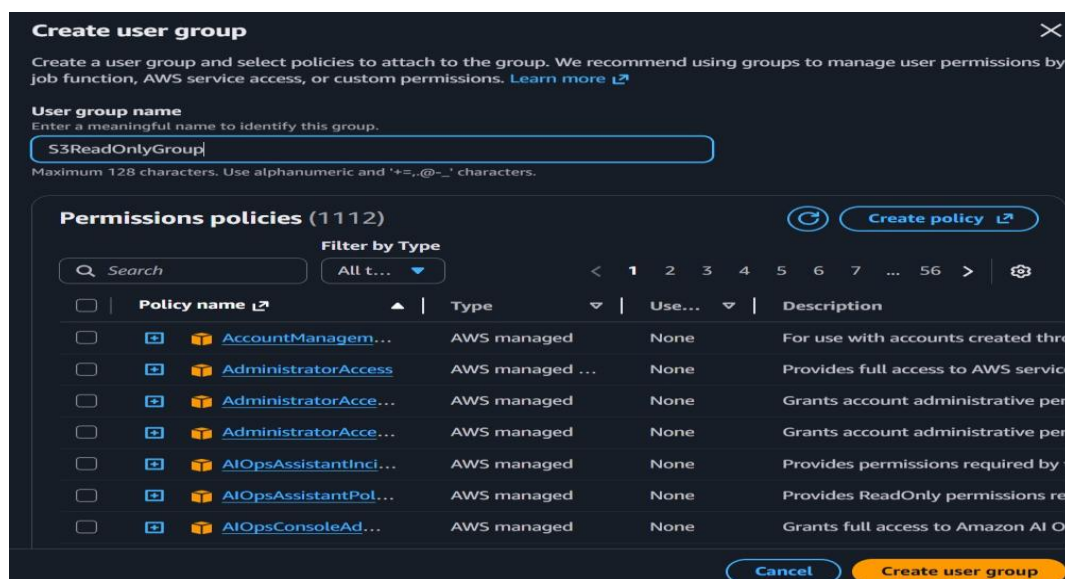
Step 4: Create a Group & Attach Policy

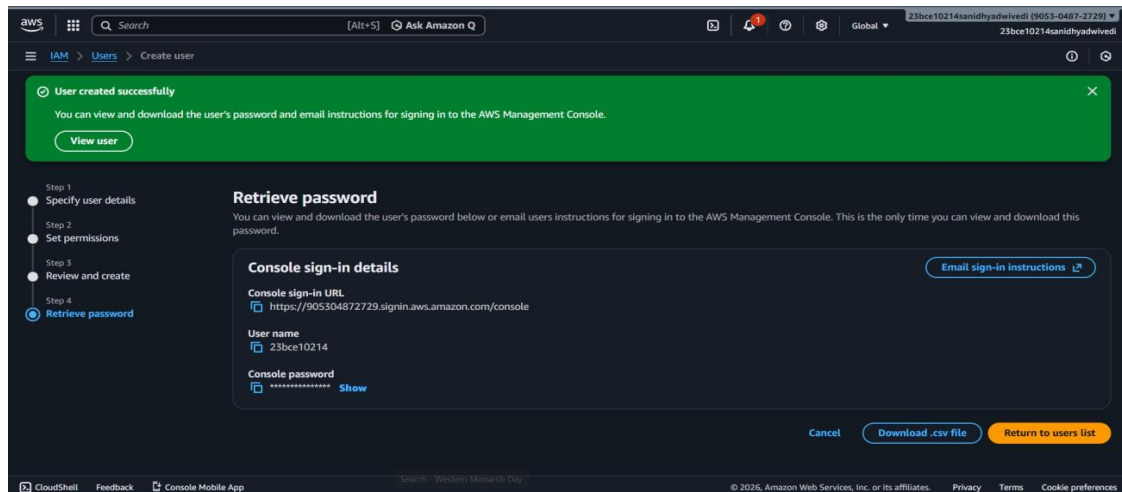
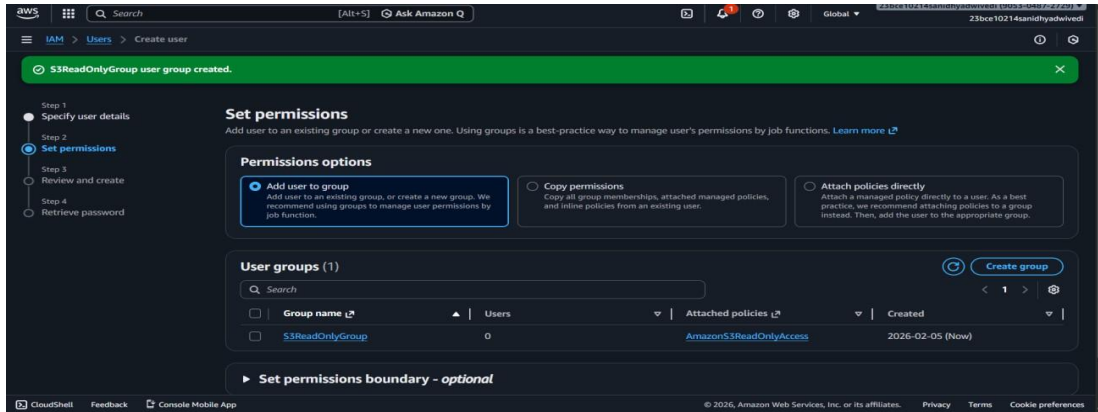
► Create Group

1. Click **Create group**
2. Group name: **S3ReadOnlyGroup**
3. Search and attach:
 - **AmazonS3ReadOnlyAccess**
4. Click **Create group**

► Add User to Group

1. Select **S3ReadOnlyGroup**
2. Click **Next**





Step 5: Create a Custom Policy

1. IAM → Policies → Create policy
2. Choose JSON tab
3. Now type this code:

```

json

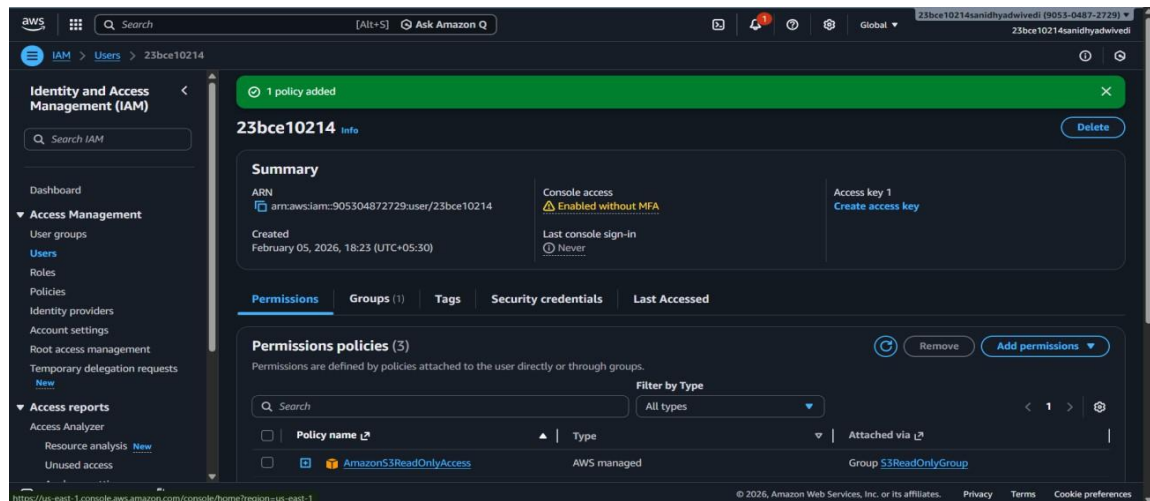
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:ListBucket",
        "s3:GetObject"
      ],
      "Resource": [
        "arn:aws:s3:::example-bucket",
        "arn:aws:s3:::example-bucket/*"
      ]
    }
  ]
}

```

The screenshot shows the AWS IAM console interface during the 'Review and create' step of creating a new policy. The breadcrumb navigation shows 'IAM > Policies > Create policy'. On the left, a progress bar indicates 'Step 1: Specify permissions' and 'Step 2: Review and create' (the current step). The main content area is titled 'Review and create' with a sub-header 'Review the permissions, specify details, and tags.' Below this, there are two main sections: 'Policy details' and 'Permissions defined in this policy'. The 'Policy details' section includes a 'Policy name' field with the value 'StudentS3BucketAccessPolicy' and a 'Description - optional' text area. The 'Permissions defined in this policy' section shows a search bar and a list of services, currently displaying 'Allow (1 of 461 services)' with a toggle to 'Show remaining 460 services'. The footer of the console shows various utility links like 'CloudShell', 'Feedback', and 'Console Mobile App', along with the copyright notice '© 2026, Amazon Web Services, Inc. or its affiliates.' and links for 'Privacy', 'Terms', and 'Cookie preferences'.

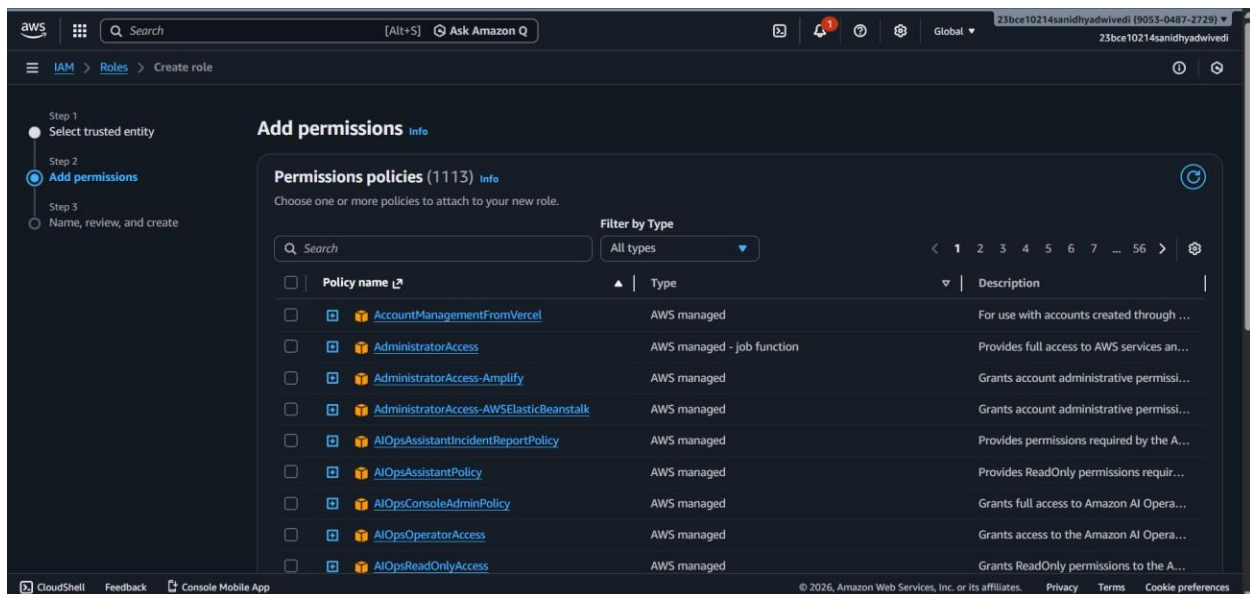
Step 6: Attach Custom Policy to User

1. IAM → **Users** → Click student-user
2. Go to **Permissions** tab → **Add permissions**
3. Choose **Attach policies directly**
4. Search → StudentS3BucketAccessPolicy
5. Select → Click **Next** → **Add permissions**



Step 7: Create a Role

1. IAM → Roles → Create role
2. Select trusted entity:
 - AWS service
3. Use case:
 - EC2
4. Click **Next**



Step 8: Attach Policy to Role

1. Search and select:
 - AmazonS3FullAccess
2. Click **Next**
3. Role name: EC2S3AccessRole
4. Click **Create role**

The screenshot shows the AWS IAM console interface during the 'Name, review, and create' step of creating a new role. The left sidebar indicates the current step is Step 3. The main content area is titled 'Name, review, and create' and contains two sections: 'Role details' and 'Trust policy'.

Role details

- Role name:** EC2S3AccessRole (Maximum 64 characters. Use alphanumeric and '+,=,@,_,.' characters.)
- Description:** Allows EC2 instances to call AWS services on your behalf. (Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-/\[\]]#%*~:~")

Step 1: Select trusted entities

Trust policy

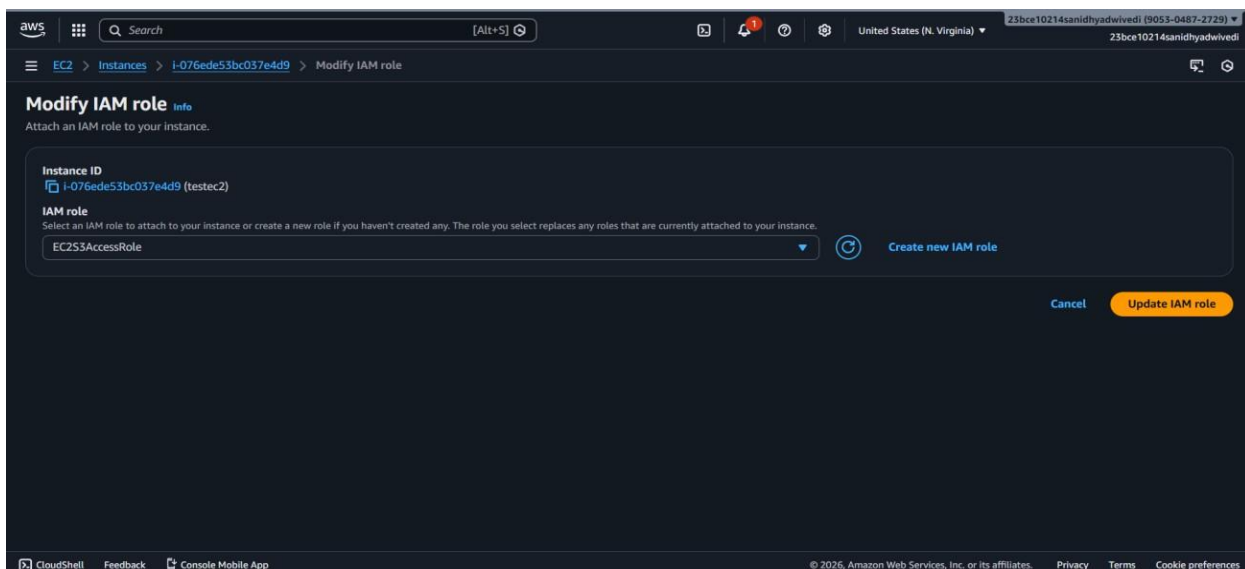
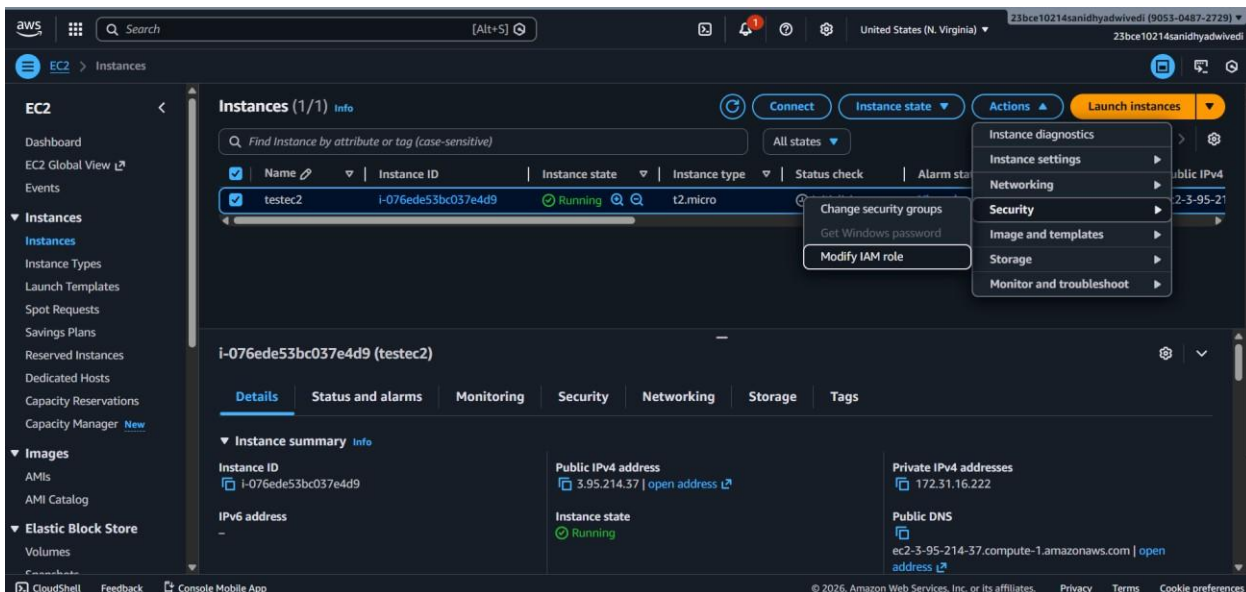
```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
```

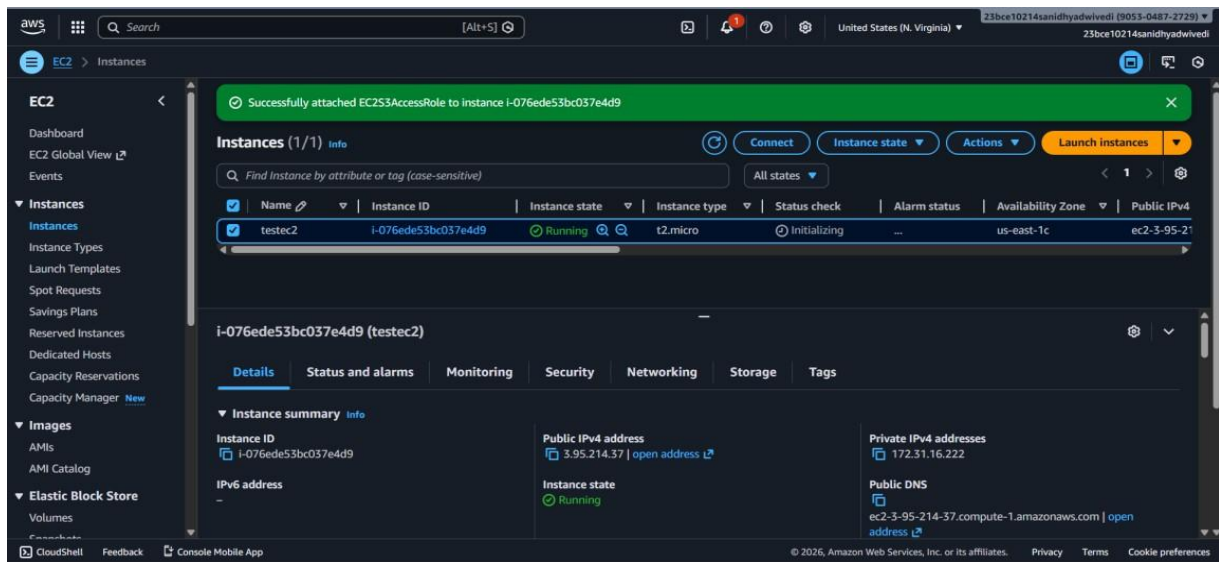
The screenshot shows the AWS IAM console interface displaying a list of roles. A green banner at the top indicates 'Role EC2S3AccessRole created.' The left sidebar shows the 'Roles' page. The main content area is titled 'Roles (10)' and includes a search bar and a table of roles.

☐	Role name	Trusted entities	Last activity
☐	aws-elasticbeanstalk-ec2-role	AWS Service: ec2	198 days ago
☐	aws-elasticbeanstalk-service-role	AWS Service: elasticbeanstalk	198 days ago
☐	AWSServiceRoleForAutoScaling	AWS Service: autoscaling (Service-Link	Yesterday
☐	AWSServiceRoleForElasticLoadBalancing	AWS Service: elasticloadbalancing (S	Yesterday
☐	AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (Se	2 hours ago
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Link	-
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service	-
☐	EC2S3AccessRole	AWS Service: ec2	-
☐	ecsTaskExecutionRole	AWS Service: ecs-tasks	-
☐	myLambdaFunction-role-70nojggl	AWS Service: lambda	198 days ago

Step 9: Attach Role to EC2 Instance

1. Go to **EC2 Dashboard**
2. Instances → Select your EC2 instance
3. Actions → Security → Modify IAM role
4. Select role: EC2S3AccessRole
5. Click **Update IAM role**





Step 10: Test Permissions

► Test User Access

1. Log out → Login using **student-user**
2. Try:
 - Open S3
 - You should be able to:
 - List bucket
 - Download objects
 - + Not upload/delete (if read-only)

